DRUGS · BLEEDING AND HAEMOSTASIS

Desmopressin

Desmopressin (DDAVP) is a synthetic vasopressin analogue that releases von Willebrand factor and factor VIII from endothelial stores, raising their levels two to five times for several hours. It is used, with haematology advice and usually with tranexamic acid, to prepare patients with type 1 von Willebrand disease, mild haemophilia A and some platelet disorders for tonsillectomy and other surgery, and to treat their bleeding. Its main danger is hyponatraemia, especially in young children.

1 CHAPTER ONE At a glance

The drug on one page.

1.1 Summary

DESMOPRESSIN

	SUMMARY
Class	Synthetic vasopressin (V2 receptor) analogue
How it works	Stimulates release of von Willebrand factor and factor VIII from endothelial Weibel-Palade bodies; also antidiuretic
ENT uses	Prophylaxis and treatment of bleeding in type 1 von Willebrand disease, mild haemophilia A and carriers, uraemic and drug-induced platelet dysfunction; tonsillectomy, adenoidectomy and other surgery in these patients
Dose	0.3 micrograms/kg intravenously or subcutaneously (maximum about 20 to 27 micrograms) over 20 to 30 minutes, 30 to 60 minutes before surgery, repeated every 12 to 24 hours for a few doses; intranasal 300 micrograms (adults) as an alternative
Effectiveness	A test dose beforehand shows whether the patient responds; tachyphylaxis after 2 to 3 doses

	SUMMARY
Avoid	Type 2B von Willebrand disease (thrombocytopenia); children under 2 years (hyponatraemic seizures); cardiovascular disease; fluid overload; habitual high fluid intake

2 CHAPTER TWO Using it in ENT

Practice.

2.1 Tonsillectomy in bleeding disorders

Screen for bleeding disorders by **history** (family history, easy bruising, prolonged bleeding after dental extraction, menorrhagia); routine clotting tests miss mild von Willebrand disease. Plan surgery with haematology: desmopressin before surgery, tranexamic acid for 7 to 10 days, overnight admission, fluid restriction and sodium checks.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

Hyponatraemia and water intoxication (fluid restriction to about two thirds of maintenance for 24 hours, sodium monitoring, particularly in children); facial flushing, headache, tachycardia, hypotension; rarely thrombosis.

3.2 Interactions

Drugs that increase antidiuretic hormone effect or cause hyponatraemia (SSRIs, carbamazepine, NSAIDs, opioids).

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Mechanism (endothelial release of von Willebrand factor and factor VIII).
- Use in **tonsillectomy** with bleeding disorders.
- **Hyponatraemia** and fluid restriction.

4.2 Common pitfalls

COMMON PITFALLS

- Using desmopressin in type 2B von Willebrand disease.
- Free intravenous fluids after desmopressin in a child.

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